

Apps of Machine Learning HW 1

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Part 1

2. Open csv file as Excel workbook.

- Since we are not to submit the whole dataset but 'open file in spreadsheet' is in rubric, this is the image of the Excel workbook I've worked with.

 .ipynb_checkpoints	2/13/2025 12:37 PM	File folder
 __pycache__	2/13/2025 10:58 PM	File folder
 diabetic_data	2/3/2025 9:02 PM	Microsoft Excel Comma Separated Values File
 diabetic_data	2/10/2025 9:48 PM	Microsoft Excel Worksheet

3. Determine the data type and role.

FIELD	ROLE	TYPE
encounter_id	ID	numeric
patient_nbr	ID	numeric
race	feature	categorical
gender	feature	categorical
age	feature	ordinal
weight	feature	ordinal
admission_type_id	feature	categorical
discharge_disposition_id	feature	categorical
admission_source_id	feature	categorical
time_in_hospital	feature	numeric
payer_code	feature	categorical

medical_specialty	feature	categorical
num_lab_procedures	feature	numeric
num_procedures	feature	numeric
num_medications	feature	numeric
number_outpatient	feature	numeric
number_emergency	feature	numeric
number_inpatient	feature	numeric
diag_1	feature	categorical
diag_2	feature	categorical
diag_3	feature	categorical
number_diagnoses	feature	numeric
max_glu_serum	feature	ordinal
A1Cresult	feature	ordinal
metformin	feature	categorical
repaglinide	feature	categorical
nateglinide	feature	categorical
chlorpropamide	feature	categorical
glimepiride	feature	categorical
acetohexamide	feature	categorical
glipizide	feature	categorical
glyburide	feature	categorical
tolbutamide	feature	categorical
pioglitazone	feature	categorical
rosiglitazone	feature	categorical
acarbose	feature	categorical

miglitol	feature	categorical
troglitazone	feature	categorical
tolazamide	feature	categorical
examide	feature	categorical
citoglipton	feature	categorical
insulin	feature	categorical
glyburide-metformin	feature	categorical
glipizide-metformin	feature	categorical
glimepiride-pioglitazone	feature	categorical
metformin-rosiglitazone	feature	categorical
metformin-pioglitazone	feature	categorical
change	feature	categorical
diabetesMed	feature	categorical
readmitted	feature	categorical

4. How did you define impossibly large values and which values did you replace them with?
- I found 5 values that have "9999999" as values in num_medications. I defined them as impossibly large since it is too much of a dose for medications. I replaced them with a blank space.

5. Univariate descriptive statistics for all of the appropriate columns in the dataset.

FIELD	enc ount er_id	patie nt_n br	r a c e	g e n d er	a g e	w e i g h t	admis sion_t ype_id	discharg e_disposi tion_id	admissi on_sou rce_id	time_i n_hos pital	pay er_ cod e	medic al_sp ecialty
Mean	N/A	N/A	N / A	N / A	N / A	N / A	2.0240 06053	3.715641 766	5.7544 36649	4.395 9868 72	N/A	N/A

Min	125 22	135	N / A	N /A	N / A	N /A	1	1	1	1	N/A	N/A
Max	443 867 222	1895 0261 9	N / A	N /A	N / A	N /A	8	28	25	14	N/A	N/A
Range	443 854 700	1895 0248 4	N / A	N /A	N / A	N /A	7	27	24	13	N/A	N/A
Median	152 388 987	4550 5143	N / A	N /A	N / A	N /A	1	1	7	4	N/A	N/A
Mode	N/A	8878 5891	N / A	N /A	N / A	N /A	1	1	7	3	N/A	N/A
Varianc e	1.05 35E +16	1.49 741E +15	N / A	N /A	N / A	N /A	2.0891 8934	27.88014 781	16.516 75303	8.910 8683 83	N/A	N/A
Std deviatio n	102 640 296	3869 6359 .35	N / A	N /A	N / A	N /A	1.4454 0283	5.280165 509	4.0640 80834	2.985 1077 67	N/A	N/A
Quartile 1	849 611 94	2341 3221	N / A	N /A	N / A	N /A	1	1	1	2	N/A	N/A
Quartile 2	152 388 987	4550 5143	N / A	N /A	N / A	N /A	1	1	7	4	N/A	N/A
Quartile 3	230 270 888	8754 5949 .75	N / A	N /A	N / A	N /A	3	4	7	6	N/A	N/A
# Missing	0	0	2 2 7 3	0	0	9 8 5 6 9	0	0	0	0	402 56	0

# zeros	0	0	0	0	0	0	0	0	0	0	0	0
# question marks	0	0	2273	0	0	98569	0	0	0	0	40256	49949

FIELD	num_lab_procedures	num_procedures	num_medications	number_outpatient	number_emergency	number_inpatient	diag_1	diag_2	diag_3	number_diagnoses	max_glu_serum	A1C_result
Mean	43.09564098	1.339657698	16.0212766	0.369357153	0.197836212	0.635565906	493.5830703	438.6748832	418.1792854	7.422606765	N/A	N/A
Min	1	0	1	0	0	0	3	5	3	1	N/A	N/A
Max	132	6	81	42	76	21	999	999	999	16	N/A	N/A
Range	131	6	80	42	76	21	996	994	996	15	N/A	N/A
Median	44	1	15	0	0	0	440	425	403	8	N/A	N/A
Mode	1	0	13	0	0	0	428	276	250	9	N/A	N/A
Variance	387.0805299	2.909375424	66.04661179	1.605960825	0.865778642	1.594823689	42731.2237	32005.79591	31443.7934	3.738809521	N/A	N/A
Std deviation	19.67436225	1.705689135	8.126906656	1.267265097	0.930472268	1.26286329	206.7153204	178.9016375	177.3239786	1.933600145	N/A	N/A

Variance	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Std deviation	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Quartile 1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Quartile 2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Quartile 3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
# Missing	0	0	0	0	0	0	0	0	0	0	0	0
# zeros	0	0	0	0	0	0	0	0	0	0	0	0
# question marks	0	0	0	0	0	0	0	0	0	0	0	0

FIELD	m i g l i t o l	tro glit az on e	tol az a mi de	e x a m i d e	cit og lip to n	i n s u l i n	glybur ide-m etfor min	glipizi de-m etfor min	glimepi ride-pi oglitaz one	metfor min-ro siglitaz one	metfor min-pi oglitaz one	c h a n g e	dia bet es Me d	re ad mi tte d
Mean	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Min	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Max	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Range	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A

Median	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Mode	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Varianc e	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Std deviat ion	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Quartil e 1	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Quartil e 2	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
Quartil e 3	N / A	N/ A	N/ A	N / A	N/ A	N / A	N/A	N/A	N/A	N/A	N/A	N / A	N/ A	N/ A
# Missing	0	0	0	0	0	0	0	0	0	0	0	0	0	0
# zeros	0	0	0	0	0	0	0	0	0	0	0	0	0	0
# questio n marks	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Part 2

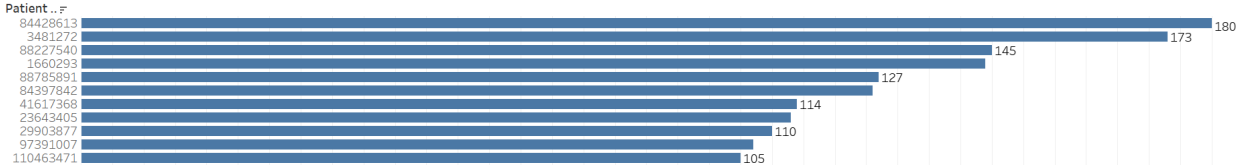
7

- a. Three patients spent the most time in hospital and the number of encounters.

Patient Nbr	Time In Hospital	Encounter Count
84428613	180	22
3481272	173	18
88227540	145	23

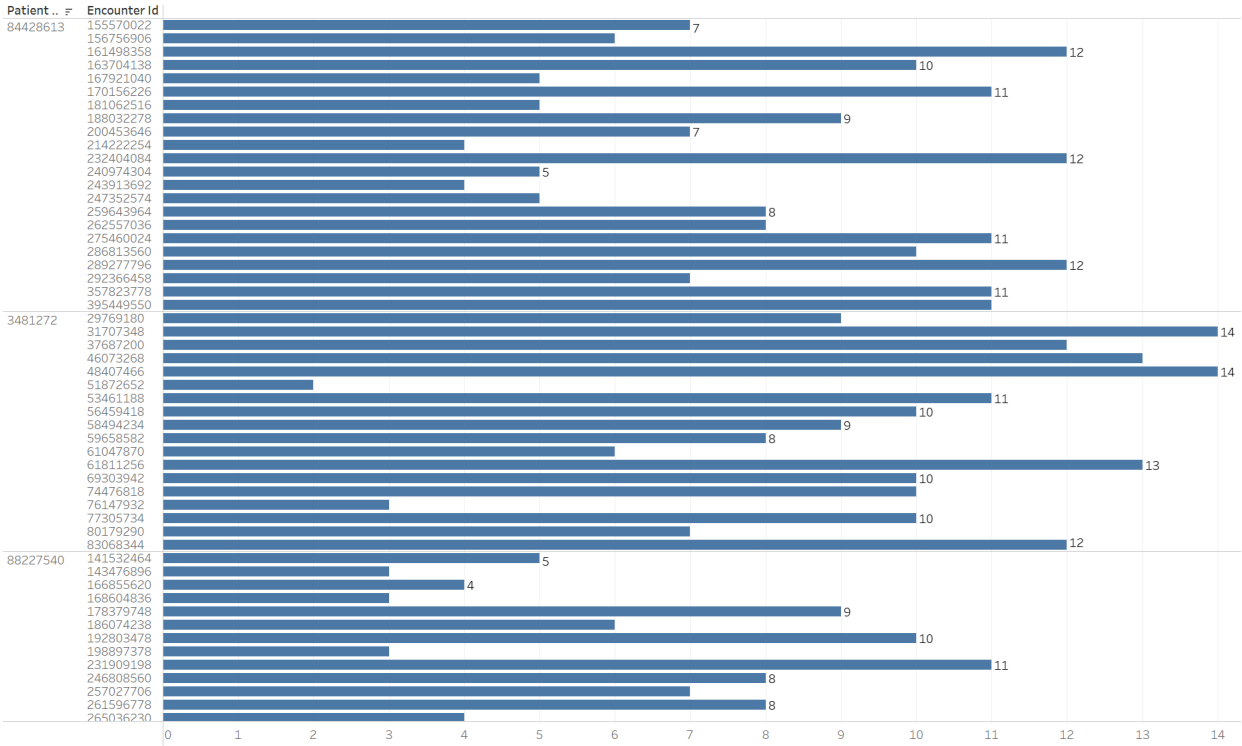
Columns	SUM(Time In Hospital)
Rows	Patient Nbr

Sheet 5

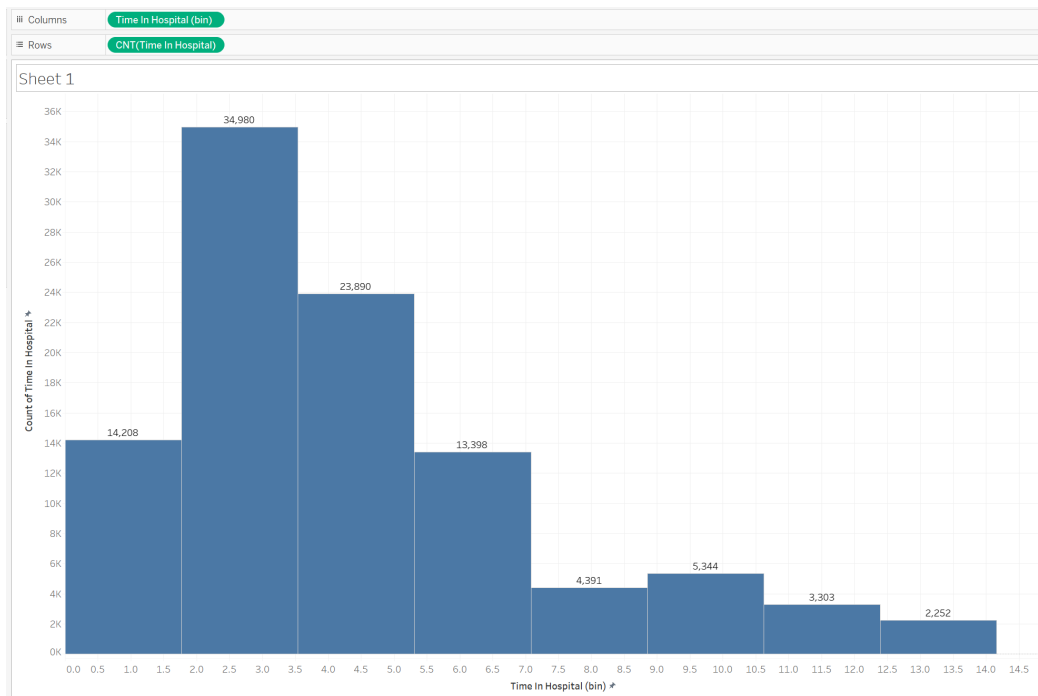


Columns	SUM(Time In Hospital)
Rows	Patient Nbr Encounter Id

Sheet 5



b. Histogram of the number of occurrences of each length of hospital stay.



c. Percentage breakdown among the different A1C results.

Pages

Filters

Marks

Automatic

Color

Size

T

Text

Detail

Tooltip

CNT(Patient ..

iii Columns

Race

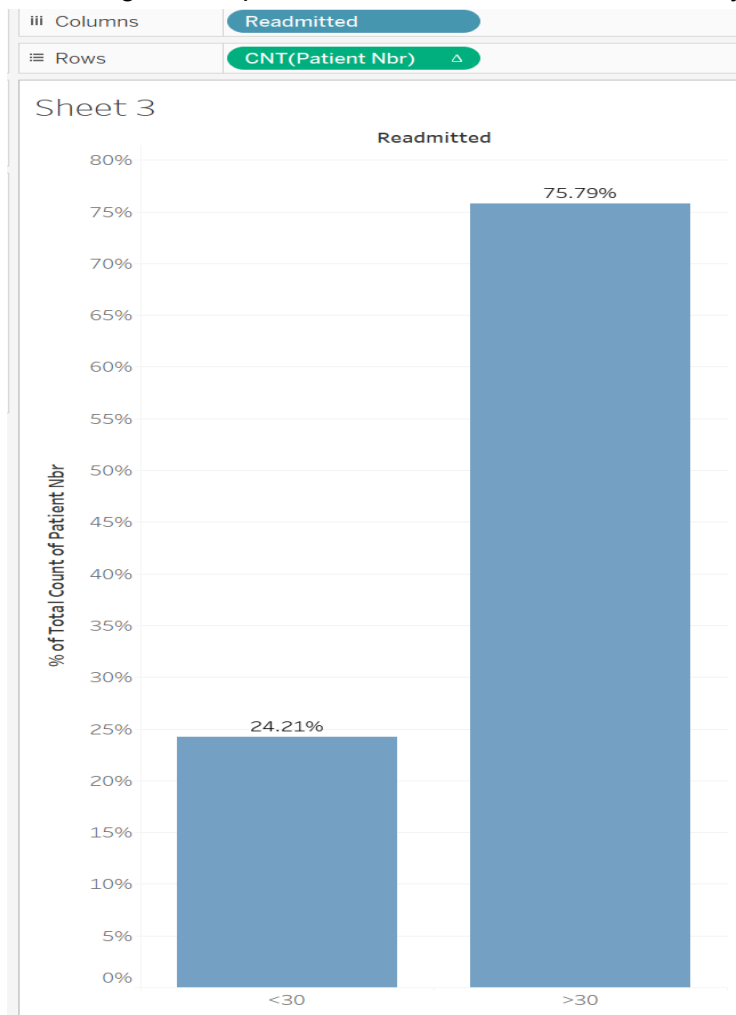
Rows

A1Cresult

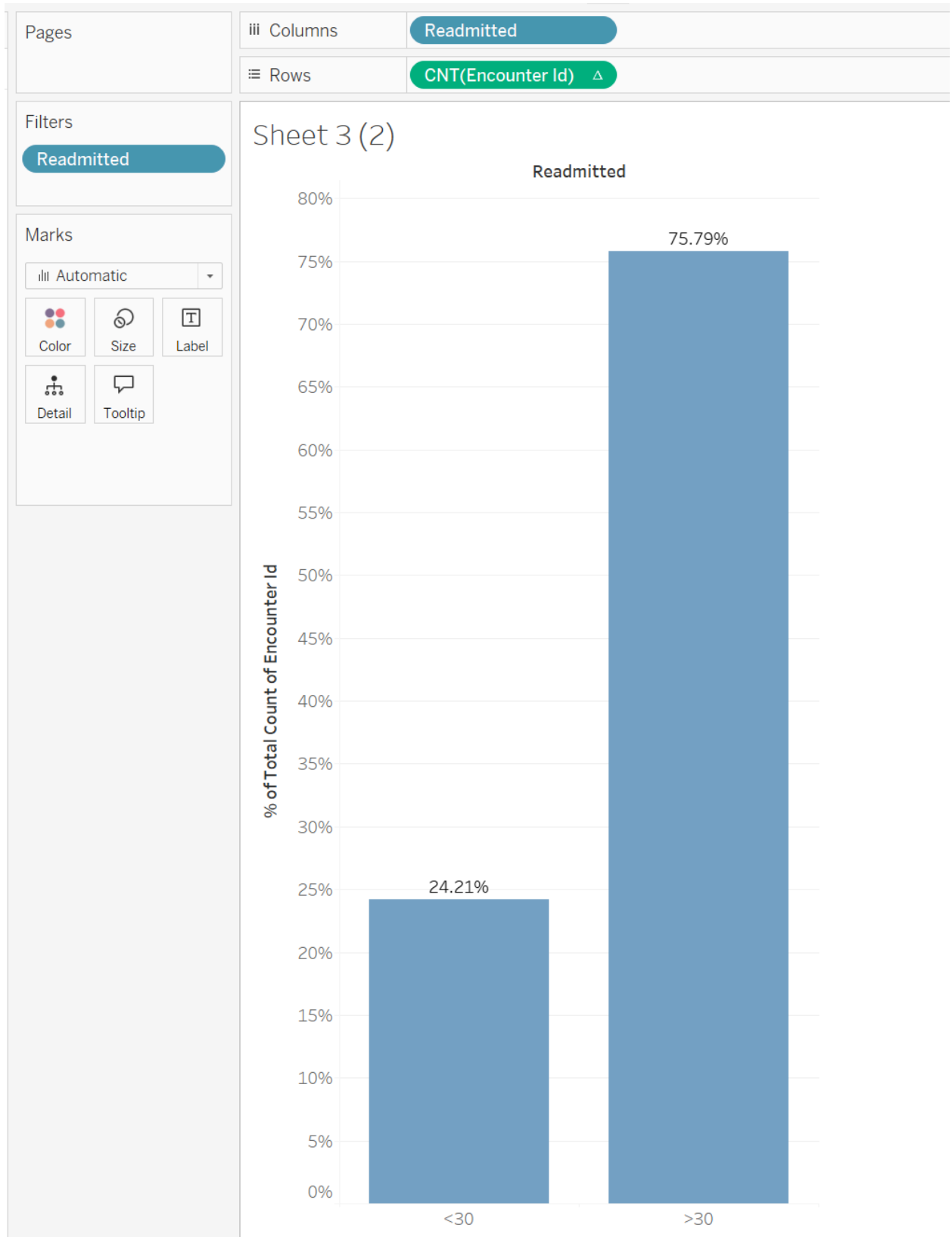
Sheet 2

	Race					
A1C...		? African..	Asian	Caucas..	Hispanic	Other
>7	3.61%	2.91%	5.62%	3.90%	4.81%	4.65%
>8	11.22%	9.60%	9.67%	7.41%	13.01%	9.96%
None	81.43%	81.73%	78.94%	84.03%	75.90%	79.75%
Norm	3.74%	5.76%	5.77%	4.66%	6.28%	5.64%

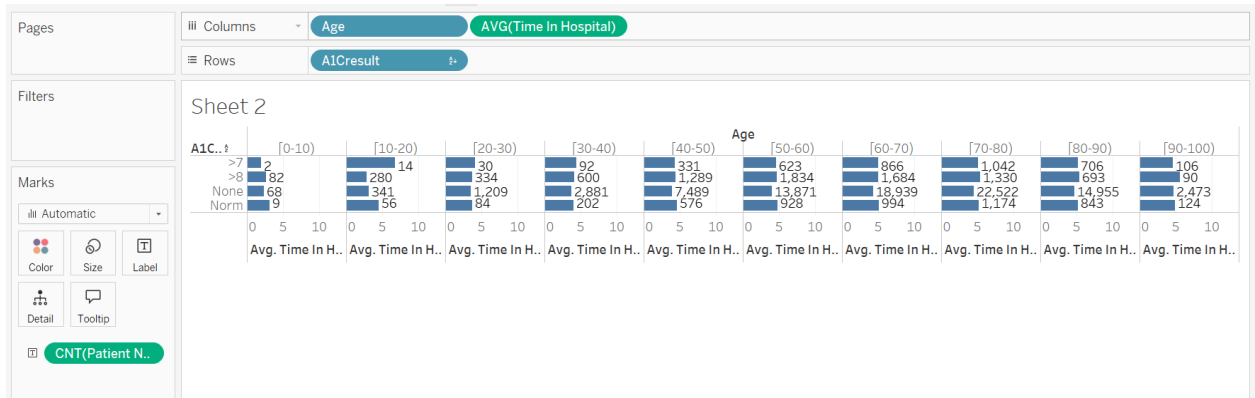
d. Percentage of the patients readmitted in less than 30 days, and in more than 30 days.



- e. Percentage of encounters resulted in a readmission in less than 30 days, and in more than 30 days?



- f. Table showing the average length of time in hospital for each different A1C result, broken down by age group.



Part 3

8. Simple Python program to load data from the spreadsheet into a Pandas dataframe.

- CODE

```
import pandas as pd
import numpy as np
filename = "diabetic_data_submit.xlsx"
#"diabetic_data_test.xlsx"
df = pd.read_excel(filename)
df.shape
```

- RESULT

(101766, 50)

9. Data quality report submission

- Data quality report is submitted with a separate file.

10. Paste all the Python code in submission.

- Codes in main.ipynb

```
import pandas as pd
import numpy as np
filename = "diabetic_data_submit.xlsx"
df = pd.read_excel(filename)
df.shape

df.head()
```

```

from StatsReport import StatsReport
myreport = StatsReport()

df =
df.drop(['race','gender','age','weight','payer_code','medical_specialty','diag_1','diag_2','diag_3','max_glu_serum',
        'A1Cresult', 'metformin', 'repaglinide',
        'nateglinide', 'chlorpropamide', 'glimepiride', 'acetohexamide',
        'glipizide', 'glyburide', 'tolbutamide', 'pioglitazone', 'rosiglitazone',
        'acarbose', 'miglitol', 'troglitazone', 'tolazamide', 'examide',
        'citoglipton', 'insulin', 'glyburide-metformin', 'glipizide-metformin',
        'glimepiride-pioglitazone', 'metformin-rosiglitazone',
        'metformin-pioglitazone', 'change', 'diabetesMed', 'readmitted'
], axis=1)
labels = df.columns

myreport.addCol(labels,df[labels])

myreport.writetofile(r"C:\dev\course_material\ece_5464_apps_of_ml\ece-5464\project_1\data_quality_report.xlsx",myreport.statsdf)

```

- Codes in StatsReport.py

```

import pandas as pd
class StatsReport:
    def __init__(self):
        self.statsdf = pd.DataFrame()
        self.statsdf['stat'] = ['types','cardinality','mean','median','stdev', 'min',
        'max','nzero','nques','nmissing']
        pass
    def addCol(self, label, d):
        zero_count = (d == 0).sum()
        question_count = (d == "?").sum()
        print(question_count)
        missing_count = (d == " ").sum()
        self.statsdf[label] = [d.dtypes,d.nunique(),d.mean(),d.median(), d.std(),d.min(),
        d.max(),zero_count,question_count,missing_count]
    def to_string(self):
        return self.statsdf.to_string()
    def writetofile(self, file_path,data):
        df = pd.DataFrame(data)
        df.to_excel(file_path, index=False)

```

class StatsReport referred from ECE5464 SP25 4

writetofile function, zero_count referred from Copilot, modified by Joong Hyun An